

Assignment: Module 2 - The LP Model

Directions

1. Back Savers is a company that produces backpacks primarily for students. They are considering offering some combination of two different models—the Collegiate and the Mini. Both are made out of the same rip-resistant nylon fabric. Back Savers has a long-term contract with a supplier of the nylon and receives a 5000 square-foot shipment of the material each week. Each Collegiate requires 3 square feet while each Mini requires 2 square feet. The sales forecasts indicate that at most 1000 Collegiates and 1200 Minis can be sold per week. Each Collegiate requires 45 minutes of labor to produce and generates a unit profit of \$32. Each Mini requires 40 minutes of labor and generates a unit profit of \$24. Back Savers has 35 laborers that each provides 40 hours of labor per week. Management wishes to know what quantity of each type of backpack to produce per week.

- a. Clearly define the decision variables

Decision Variables:

- Collegiate Backpacks Produced per week: X_1
- Mini Backpacks Produced per week: X_2

- b. What is the objective function?

Objective Function:

- Maximize Profit

- c. What are the constraints?

Constraints	
Nylon Supply	$3(X_1) + 2(X_2) \leq 5000$
Labor Constraint	$.75(X_1) + .6667(X_2) \leq 1400$
Collegiate Demand	$X_1 \leq 1000$
Mini Demand	$X_2 \leq 1200$
Non-negativity	$X_1, X_2 \geq 0$

- d. Write down the full mathematical formulation for this LP problem.

- Maximize $(Z) = 32(X_1) + 24(X_2)$
- Subject to:
- $3(X_1) + 2(X_2) \leq 5000$
- $.75(X_1) + .6667(X_2) \leq 1400$
- $X_1 \leq 1000$
- $X_2 \leq 1200$
- $X_1, X_2 \geq 0$

2. The Weigelt Corporation has three branch plants with excess production capacity. Fortunately, the corporation has a new product ready to begin production, and all three plants have this capability, so some of the excess capacity can be used in this way. This product can be made in three sizes--large, medium, and small--that yield a net unit profit of \$420, \$360, and \$300, respectively. Plants 1, 2, and 3 have the excess capacity to produce 750, 900, and 450 units per day of this product, respectively, regardless of the size or combination of sizes involved.

The amount of available in-process storage space also imposes a limitation on the production rates of the new product. Plants 1, 2, and 3 have 13,000, 12,000, and 5,000

square feet, respectively, of in-process storage space available for a day's production of this product. Each unit of the large, medium, and small sizes produced per day requires 20, 15, and 12 square feet, respectively.

Sales forecasts indicate that if available, 900, 1,200, and 750 units of the large, medium, and small sizes, respectively, would be sold per day.

At each plant, some employees will need to be laid off unless most of the plant's excess production capacity can be used to produce the new product. To avoid layoffs if possible, management has decided that the plants should use the same percentage of their excess capacity to produce the new product.

Management wishes to know how much of each of the sizes should be produced by each of the plants to maximize profit.

- a. Define the decision variable

Decision Variables	
L1	Quantity of Large Produced @ Plant 1
L2	Quantity of Large Produced @Plant 2
L3	Quantity of Large Produced@ Plant 3
M1	Quantity of Medium produced @ Plant 1
M2	Quantity of Medium produced @ Plant 2
M3	Quantity of Medium produced @ Plant 3
S1	Quantity of Small Produced @ Plant 1
S2	Quantity of Small Produced @ Plant 2
S3	Quantity of Small Produced @ Plant 3

	Plant 1	Plant 2	Plant 3
Capacity	750	900	450
Storage	13000	12000	5000

	Large	Medium	Small
Profit	420	360	300
Storage Required	20	15	12
Demand	900	1200	750

b. Formulate a linear programming model for this problem

- Max profit (Z) = $420(L1 + L2 + L3) + 360(M1 + M2 + M3) + 300(S1 + S2 + S3)$
- Subject to:

Demand Large	$L1 + L2 + L3 \leq 900$
Demand Medium	$M1 + M2 + M3 \leq 1200$
Demand Small	$S1 + S2 + S3 \leq 750$
Capacity Plant 1	$L1 + M1 + S1 \leq 750$
Capacity Plant 2	$L2 + M2 + S2 \leq 900$
Capacity Plant 3	$L3 + M3 + S3 \leq 450$
Storage Room Plant 1	$20L1 + 15M1 + 12S1 \leq 13000$
Storage Room Plant 2	$20L2 + 15M2 + 12S2 \leq 12000$
Storage Room Plant 3	$20L3 + 15M3 + 12S3 \leq 5000$
Plant 1 percentage constraint	$L1 + M1 + S1 = 750P$
Plant 2 percentage constraint	$L2 + M2 + S2 = 900P$
Plant 3 percentage constraint	$L3 + M3 + S3 = 450P$
P = Proportion of excess capacity used at each plant	
$0 \leq P \leq 1$	
Non Negativity	$L1, L2, L3, M1, M2, M3, S1, S2, S3 \geq 0$